

# The UK Co-Benefits Atlas:

## An Interactive Visualisation Atlas to Understand the Impacts of Achieving Climate Action Targets

The UK Co-Benefits Atlas is an interactive online visualisation atlas [1] presenting and explaining data on the potential socio-economic impacts of achieving climate action targets in the UK.

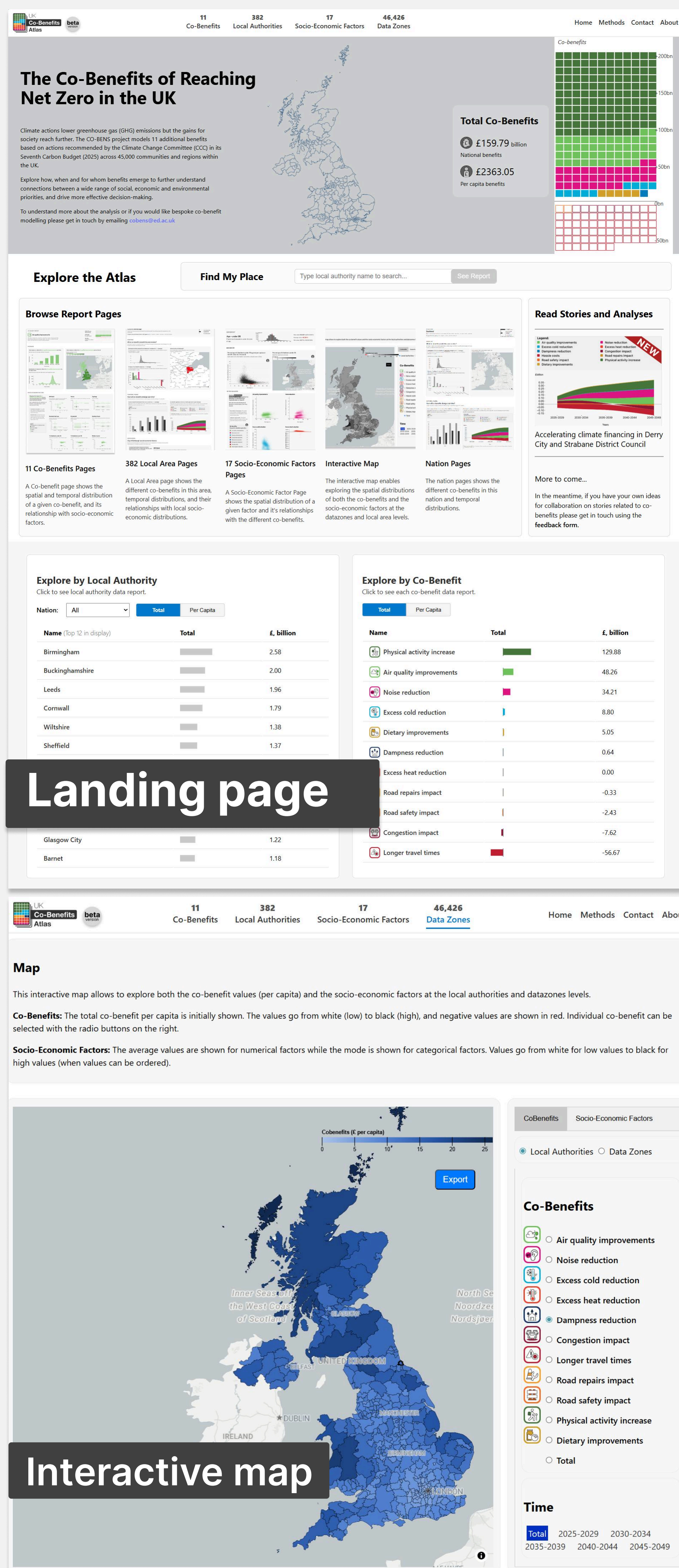
### What Are Co-Benefits?

For every £1 spent on climate change mitigation in the UK, there are as much as £14 of social benefits in the form of improved public health, better urban connectivity, and increased productivity [2].

### What is in the Co-benefits Atlas?

The Atlas integrates 410 report pages spanning 11 co-benefit categories, 17 socio-economic factors, 382 geographic reports at both national and local authority levels, and an interactive map displaying co-benefit data across 46,000 data zones.

## Atlas Pages



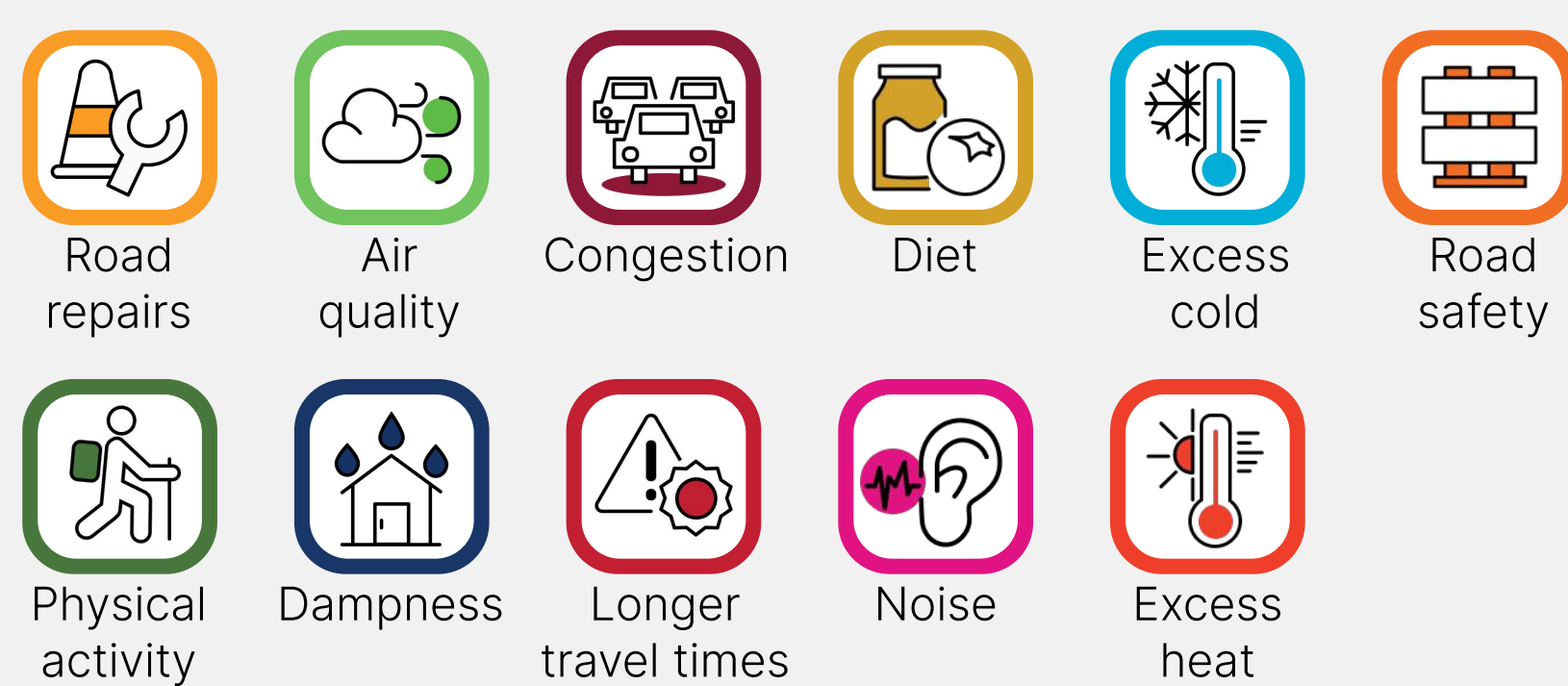
## Design Challenges and Solutions

The Atlas is designed through domain expert collaboration and stakeholder participation over a 10-month period.

The Atlas uses a **bottom-up, data-driven** approach to provide balanced access to the full range of co-benefits data.

**Transparency is maximised** by giving users options to explore different data representation, e.g., toggling between per capita values and total values.

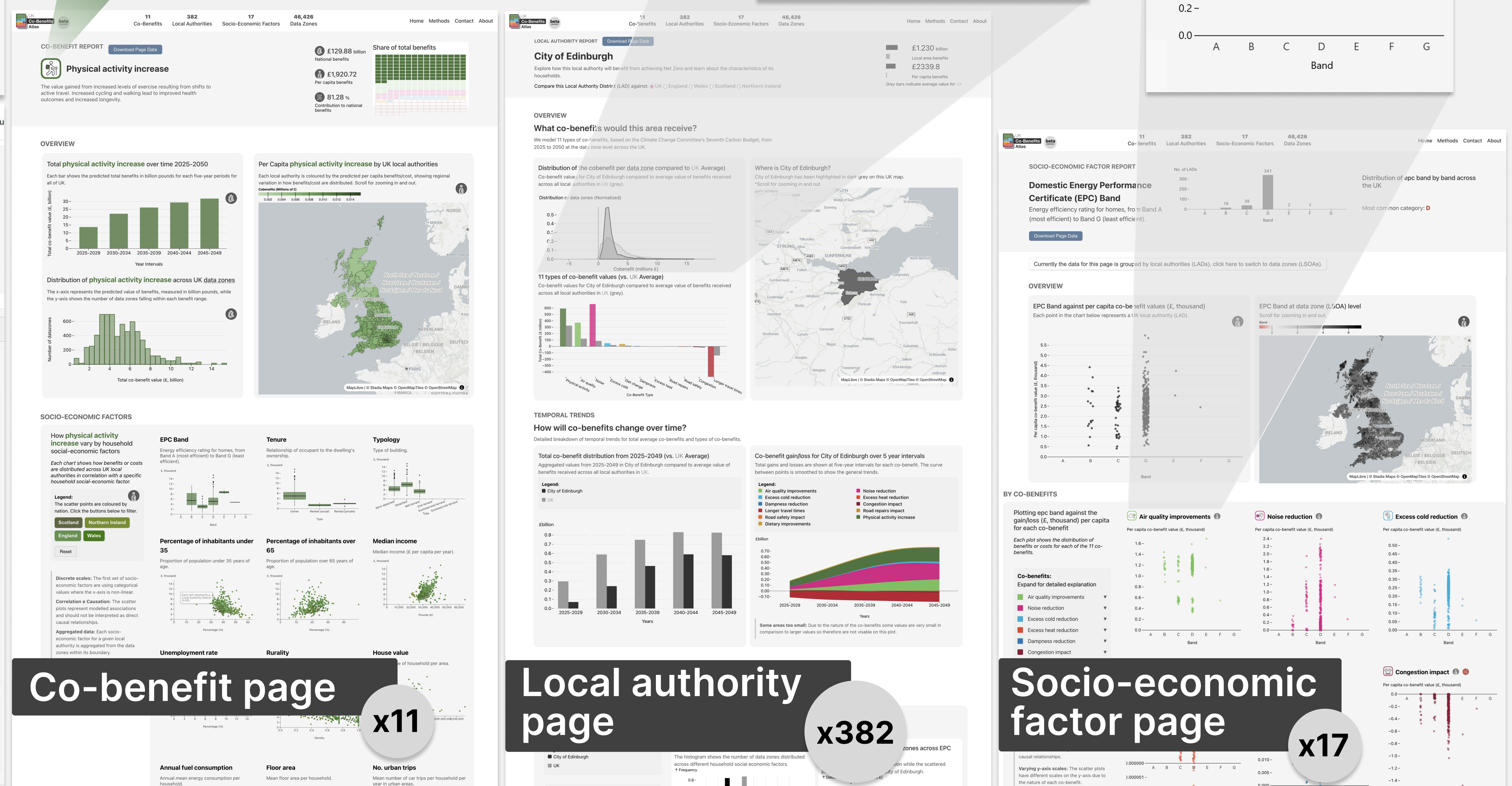
Strong **visual identities** ensure easy orientation by assigning every co-benefit a unique colour and icon.



Navigation is **coherent and intuitive**, with visual marks linking pages to foster free exploration.

Every visualisation is designed to **stand alone** with a brief explanation of its visual encoding, measurement units and label descriptions for comprehensive exports.

The use of **standard visualisations** allows exploration of the expressive limits of traditional charts while drawing on widely familiar forms.



### Potential Use Cases

Support learning, community engagement, and decision-making by providing evidence of co-benefits that help motivate action, raise awareness, and strategically guide future climate initiatives.

### Wide Audience

Aimed at policy experts and decision makers, businesses, climate activists and advocates, local communities, and the general public to explore and build their case for specific local entity.

#### References

- [1] J. Wang, X. Shu, B. Bach, and U. Hinrichs. Visualization atlases: Explaining and exploring complex topics through data, visualization, and narration. *IEEE TVCG*, 2024. 1, 2
- [2] A. Sudmant, D. Boyle, R. Higgins-Lavery, A. Gouldson, A. Boyle, J. Fulker, and J. Brogan. Climate policy as social policy? a comprehensive assessment of the economic impact of climate action in the uk. *Journal of Environmental Studies and Sciences*, 2024.

<https://ukcobenefitsatlas.net/>

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